Namrata Singh

Orai, Uttar Pradesh
+91-8853547733 nmrta1902@gmail.com GitHub

OBJECTIVE

Research-oriented M.Tech student interested in AI, machine learning, deep learning, cybersecurity, seeking research opportunities in intelligent healthcare and sensor-based systems.

EDUCATION

Maulana Azad National Institute of Technology (MANIT) 2024 – 2026
M.Tech in Computer Science and Engineering — CGPA: 8.70 *Bhopal, India*

Dr. A.P.J. Abdul Kalam Technical University 2019 – 2023
B.Tech in Computer Science and Engineering — CGPA: 7.97 *Lucknow, India*

ONGOING RESEARCH

Gait Analysis Using Pressure Insoles and IMU Sensors
Working on gait speed prediction and walking pattern analysis using pressure insole data and IMU sensor signals for healthcare and rehabilitation applications.

PUBLICATIONS

- **Machine Learning-Based Gait Speed Classification via Multi-Model Data Fusion for Medical Disease Analysis** – Accepted at ICIoTCT 2025.
- **DFast: A CPU-Efficient and Calibrated Pipeline for Static Malware Detection** – Accepted at RCSM 2025.
- **A Spatiotemporal Deep Learning Architecture for Gait Speed Classification Using Depth-Based Pose Sequences** – Submitted to IEEE Sensors Journal (Under Review).

PROJECTS

Deepshield: AI-Powered Deepfake Detection System
– Developed a CNN-RNN based deepfake detection system for identifying manipulated videos.

Wi-Fi Security Assessment Tool
– Built a tool for Wi-Fi vulnerability assessment and security analysis.

Movie Recommendation System
– Developed a recommendation system using user preference and similarity analysis.

INTERNSHIPS

Python with Django Framework 45 Days
Training Internship

Software Development Intern July 2023 – Jan 2024
KodNest Technologies Pvt Ltd

TECHNICAL SKILLS

Languages: Python, Java

Frameworks/Tools: TensorFlow, Django, Spring Boot, GitHub

Hardware & Tools: Intel RealSense Depth Camera, ESP32, Plantar Pressure Sensors, IMU Sensors

Database: MySQL, Oracle

ACHIEVEMENTS

- Won 2nd prize in Cyber Safety Hackathon 2025 organized by Bhopal Police for Deepshield project.
- Runner-Up at Innovate 2025 Smart Hackathon for Privacy Pulse cybersecurity platform.